

Managerial Economics

Master-Level Revision Notes — MBA / PGDM

1. Market Foundations & Production Economics

Core Concept

- Production function: $Q = f(L, K)$ — output as a function of labor (L) and capital (K).
- **Short run (SR)**: at least one input fixed (usually K). **Long run (LR)**: all inputs variable.
- **Isoquant**: locus of (L, K) combinations producing the same output Q. Convex to origin because inputs are imperfect substitutes.
- **MRTS**: rate at which K can be given up per extra unit of L while holding Q constant — the slope of the isoquant.

Decision Rules / Formulas

$$\text{MRTS}_{LK} = -dK/dL \text{ (at constant } Q) = MP_L / MP_K$$

- **Diminishing marginal returns** (SR, ONE input varies): as L rises with K fixed, MP_L eventually falls. This is a per-unit-of-variable-input concept.
- **Returns to scale** (LR, ALL inputs scaled by same factor t):
 - $f(tL, tK) > t \times f(L, K)$ -> Increasing Returns to Scale (IRS)
 - $f(tL, tK) = t \times f(L, K)$ -> Constant Returns to Scale (CRS)
 - $f(tL, tK) < t \times f(L, K)$ -> Decreasing Returns to Scale (DRS)
- **Economies/diseconomies of scale** = the LR cost mirror of returns to scale: falling LAC as Q rises = economies (driven by IRS, bulk discounts, specialization); rising LAC = diseconomies (coordination/agency costs).
- **Cost definitions**:
 - Accounting cost = explicit cash outlays only.
 - Economic cost = Explicit cost + Implicit (opportunity) cost.
 - Opportunity cost = value of the best foregone alternative.
 - Sunk cost = already incurred, irrecoverable -> irrelevant to forward decisions.
 - Fixed cost = doesn't vary with Q in SR (fixed is not sunk; e.g., rented machinery is fixed but recoverable by ending the lease).
- **Shutdown rule (SR)**: Shut down (produce zero) if $P < \min AVC$. If $AVC < P < ATC$ -> keep producing (loss-minimizing, since it covers all variable cost + part of FC, vs. losing all of FC if shut down).

Graph

Isoquants and MRTS

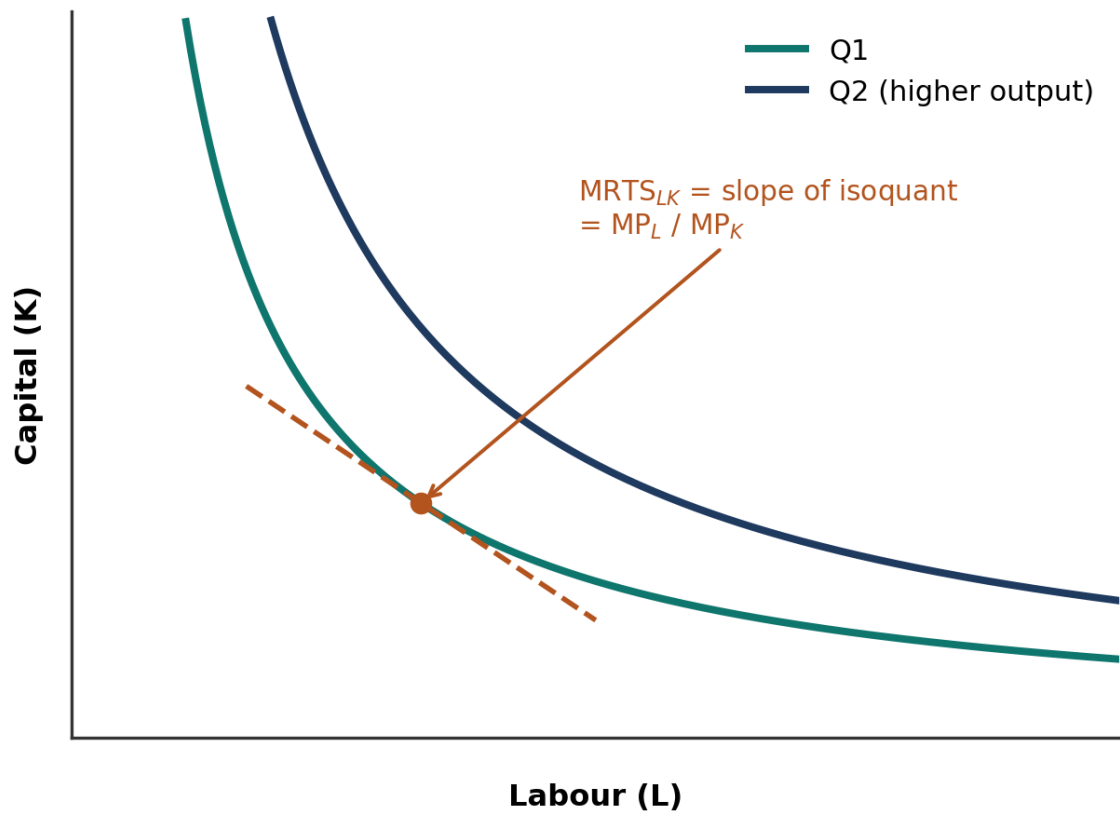


Figure 1.1 — Isoquants and MRTS

Short-Run Shutdown Rule

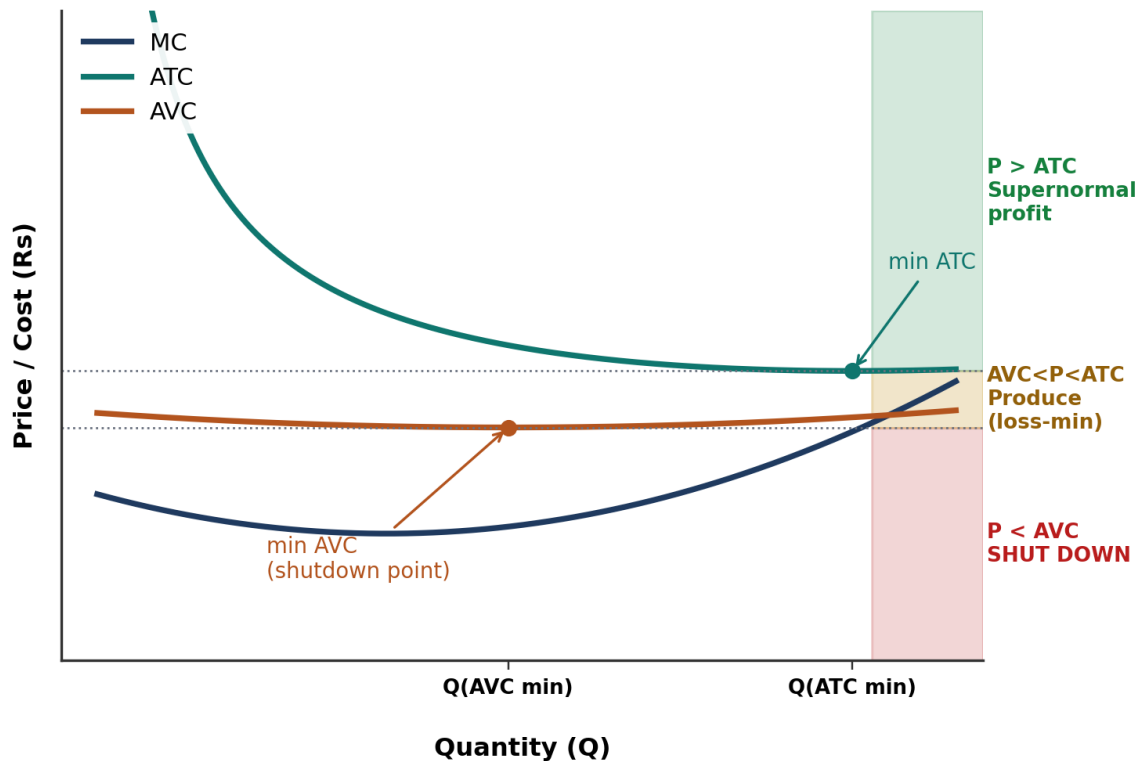


Figure 1.2 — Short-Run Shutdown Rule

Real-World Example

- **Diminishing returns:** An FMCG plant with fixed floor space keeps adding workers — output per additional worker falls past a point (floor congestion, tool-sharing delays).
- **Returns to scale:** A cloud SaaS firm doubles servers and engineers together — capacity more than doubles due to load-balancing efficiencies -> IRS.
- **Sunk cost trap:** An airline already paid a non-refundable Rs 50 cr lease deposit — that Rs 50 cr must NOT factor into the decision to continue or scrap the route.
- **Shutdown:** A textile mill's price crashes below AVC in a demand slump -> mill shuts temporarily rather than run at a bigger loss.

Quick-Fire FAQ

Q: Diminishing returns vs. diseconomies of scale?

A: SR + one input fixed (diminishing returns) vs. LR + all inputs scaled (diseconomies of scale) — different time horizons, different mechanisms.

Q: Why $P < AVC$, not $P < ATC$, for shutdown?

A: FC is sunk in the SR regardless of the decision; producing is optimal as long as price covers variable cost and contributes *something* toward FC.

Write in notes: the MRTS formula, the three ATC/AVC shutdown zones. Don't write: full isoquant derivations — just know the shape and intuition.

2. Profit Maximization Foundations

Core Concept

Profit is maximized where the additional revenue from one more unit equals the additional cost of producing it — beyond that, extra units destroy value.

Decision Rule

$$\text{Profit} = \text{TR} - \text{TC} \Rightarrow \frac{d(\text{Profit})}{dQ} = \text{MR} - \text{MC} = 0 \Rightarrow \text{MR} = \text{MC}$$

- **Second-order condition:** MC must cut MR **from below** (MC rising faster than MR at the intersection) — this is the true profit-max point, not a profit-min point.
- **Marginal logic:**
 - If $\text{MR} > \text{MC}$ → produce more (each extra unit adds more revenue than cost).
 - If $\text{MR} < \text{MC}$ → cut back (each extra unit destroys value).
 - Stop exactly where $\text{MR} = \text{MC}$.

Graph

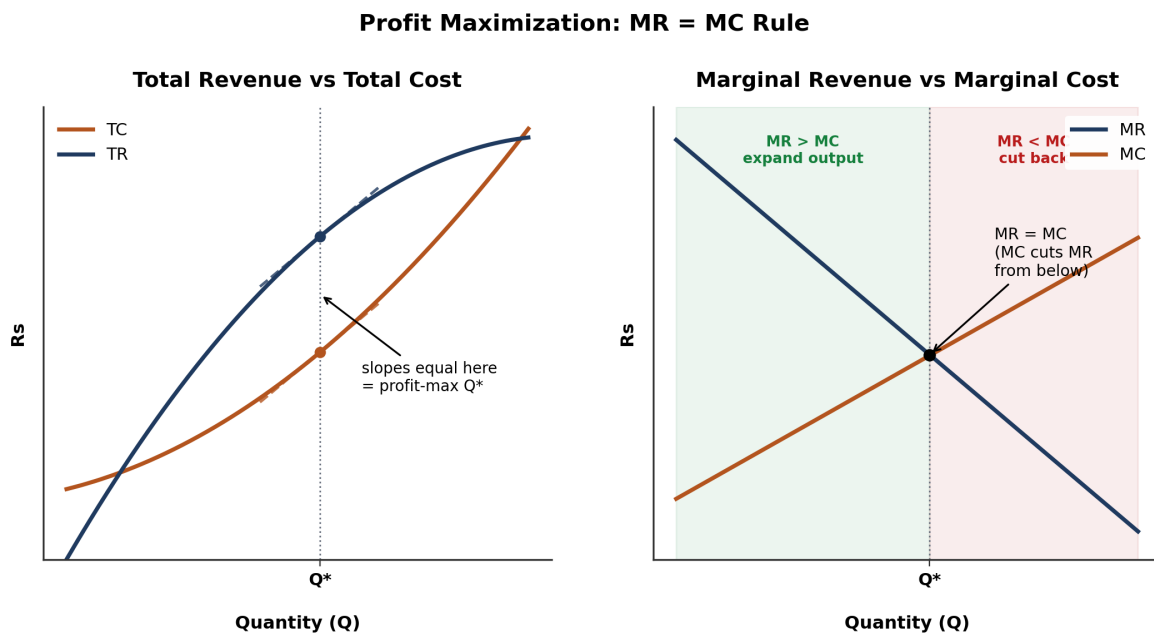


Figure 2.1 & 2.2 — Total Revenue vs Total Cost, and Marginal Revenue vs Marginal Cost

Real-World Example

A telecom operator deciding whether to add one more cell tower: if the marginal subscriber revenue it unlocks (MR) exceeds the marginal cost of that tower (MC), build it; once $\text{MR} < \text{MC}$ (saturated area), stop.

Quick-Fire FAQ

Q: Why not just maximize TR?

A: Maximizing TR ignores cost — you'd overproduce into the region where $MC > MR$ and destroy profit.

Q: Can $MR = MC$ give a loss-minimizing point instead of max profit?

A: Yes — that's why the second-order condition (MC cutting MR from below) matters; check it before declaring Q^* optimal.

Write in notes: $MR=MC$ rule + the second-order condition. Don't write: calculus derivation, just the rule and the "why."

3. Perfect Competition

Core Concept

- Assumptions: price-taking firms (no individual market power), homogeneous product, perfect information, free entry/exit.
- **Firm's demand curve** = horizontal (perfectly elastic) at market price $\rightarrow P = AR = MR$.
- **Industry demand curve** = normal downward-sloping aggregate demand.

Decision Rules

- **SR equilibrium** (firm produces where $MR=MC$, i.e., $P=MC$):
 - $P > ATC \rightarrow$ supernormal (economic) profit
 - $P = ATC \rightarrow$ normal profit (zero economic profit, but positive accounting profit)
 - $AVC < P < ATC \rightarrow$ SR loss, but firm still produces (per shutdown rule)
 - $P < AVC \rightarrow$ shut down
- **LR equilibrium:** $P = MR = LMC = \min LAC \rightarrow$ zero economic profit (entry erodes supernormal profits; exit ends losses).
- **Strategy escape hatch:** firms avoid commoditization (and the $P=MC$ /zero-profit trap) by branding/differentiating — shifting from a perfectly competitive structure toward monopolistic competition.

Graph

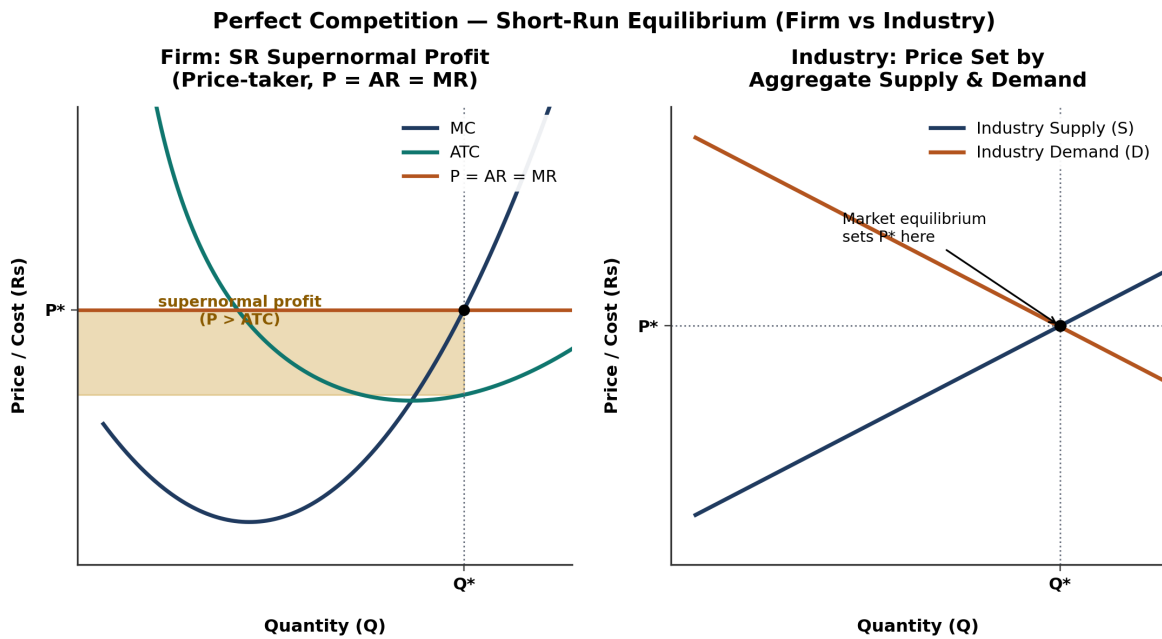


Figure 3.1 — Short-Run Equilibrium: Firm (Supernormal Profit) vs Industry

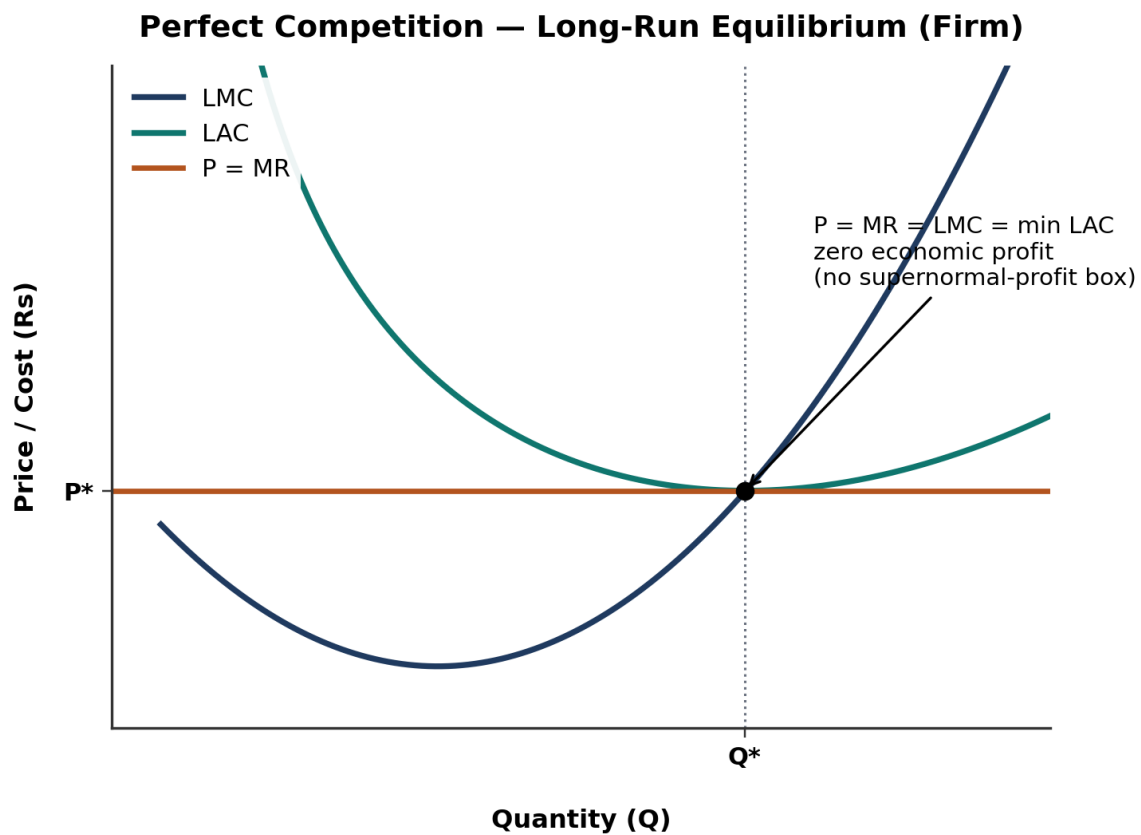


Figure 3.2 — Long-Run Equilibrium (Zero Economic Profit)

Real-World Example

Loose/unbranded staples (raw wheat, unbranded pulses in a mandi) behave close to perfect competition — thin margins, price-takers.

Escape: converting loose flour into branded "Aashirvaad" atta or loose tea into a branded packaged blend — moves the firm off the $P=MC$ treadmill by creating (limited) pricing power via differentiation.

Quick-Fire FAQ

Q: Why does LR profit go to zero under perfect competition?

A: Free entry: supernormal profits attract new entrants -> industry supply shifts right -> price falls to min LAC until profit incentive disappears.

Q: Firm vs industry demand curve — why different shapes?

A: Firm is a price-taker (infinitesimally small relative to market) -> horizontal; industry aggregates all buyers -> normal downward slope.

Write in notes: the four SR profit/loss conditions with the P vs ATC/AVC comparisons — these are the most commonly tested lines.

4. Monopoly

Core Concept

Single seller with market power, protected by entry barriers: legal barriers/patents-IP, economies of scale (natural monopoly — e.g., utilities/pipelines), network effects, or control of a critical input/technology (e.g., ASML's EUV lithography dominance).

Monopolist faces the entire (downward-sloping) market demand curve directly — it IS the industry.

Decision Rule / Math

If inverse demand is $P = a - bQ$, then $TR = PQ = aQ - bQ^2$, so

$$MR = a - 2bQ$$

-> **MR has the same intercept but twice the slope of a linear demand curve.**

- Profit max: $MR = MC$ -> gives Q^* ; read P^* off the **demand** curve (not MR) at Q^* .
- **Deadweight loss (DWL):** monopolist restricts output below the competitive ($P=MC$) level and charges above MC -> allocative inefficiency; the triangle between competitive Q and monopoly Q , bounded by demand and MC, is lost surplus captured by no one.

Graph

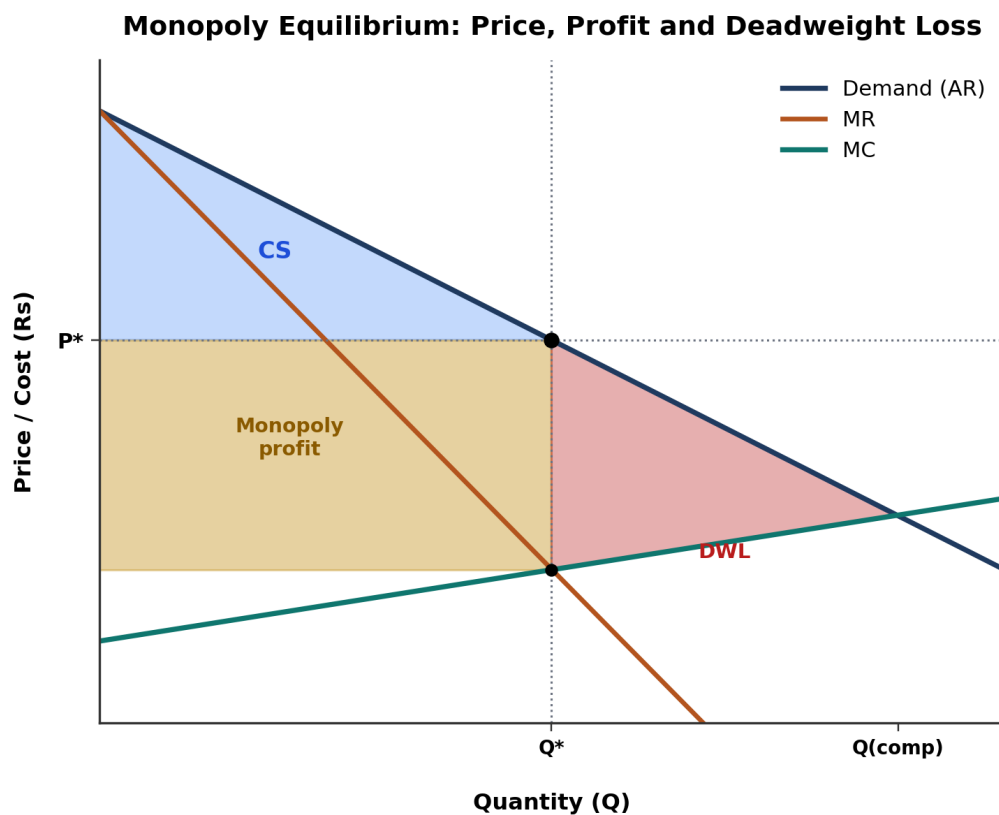


Figure 4.1 — Monopoly Equilibrium: Price, Profit and Deadweight Loss

Consumer Surplus (CS) = triangle above P^* , below demand, left of Q^* . Monopoly profit rectangle = $(P^* - ATC) \times Q^*$.
 DWL triangle = between Q^* and the competitive quantity, bounded by demand and MC.

Real-World Example

- ASML: sole global maker of EUV lithography machines needed for cutting-edge chips -> near-monopoly via technology/patent barriers, restricts output relative to competitive levels, prices high.
- Indian Railways (freight in many corridors) or a city's sole water utility = natural monopoly via massive fixed-cost economies of scale.

Quick-Fire FAQ

Q: Why does MR have twice the slope of linear demand?

A: Because $TR=PQ$ and differentiating $aQ-bQ^2$ w.r.t. Q gives $a-2bQ$ — the coefficient on Q doubles.

Q: Does a monopolist always earn supernormal profit?

A: No — only if $P > ATC$ at Q^* . It's possible (rare) for a monopolist to earn normal profit or even a loss if demand is weak relative to costs.

Write in notes: the $MR = a-2bQ$ shortcut and the DWL triangle logic — both are high-frequency numerical/graph questions.

5. Oligopoly

Core Concept

Few large firms, **mutual interdependence** (each firm's best move depends on rivals' moves — the defining feature vs. other structures), significant entry barriers, product can be homogeneous (steel, cement) or differentiated (smartphones, colas).

Market Structures Comparison

Structure	# Sellers	Product	Entry Barriers	Price Control	Example
Perfect Competition	Many	Homogeneous	None	None (price-taker)	Wheat mandi
Monopolistic Competition	Many	Differentiated	Low	Some (via branding)	Restaurants, salons
Oligopoly	Few	Homo. or Diff.	High	Significant, interdependent	Telecom, autos, cement
Monopoly	One	Unique	Very high / blocked	High	Utilities, ASML
Monopsony	One buyer	—	High (buyer side)	High (as buyer)	Single defense contractor buyer
Oligopsony	Few buyers	—	High (buyer side)	Significant (buyer side)	A few large cotton-ginning buyers facing many farmers
Cartel	Few, colluding	Homo. or Diff.	High	Jointly set like monopoly	OPEC

Decision Rule / Logic

- **Collusive:** firms jointly restrict output/fix price to mimic monopoly profit-sharing (cartel), but each member has an individual incentive to secretly cheat (over-produce) — inherently unstable without enforcement.
- **Non-collusive (price wars):** each firm assumes rivals will match price cuts but NOT match price hikes -> **kinked demand curve**.
 - Above current price: demand is elastic (rivals don't follow a hike -> you lose lots of customers).
 - Below current price: demand is inelastic (rivals match a cut -> you don't gain much volume).
- This kink creates a **discontinuous (vertical gap) MR curve** at the kink quantity -> MC can shift within that gap without changing P or Q -> explains price rigidity/"sticky prices" in oligopoly.

Graph

Oligopoly — Kinked Demand Model (Price Rigidity)

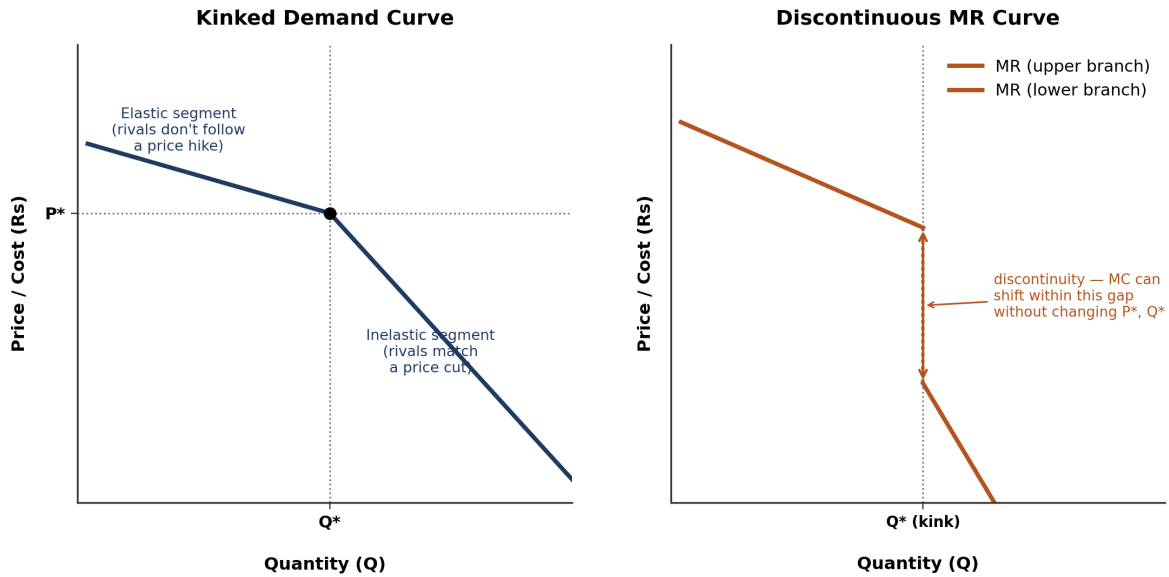


Figure 5.1 — Kinked Demand Curve and the Resulting Discontinuous MR

Real-World Example

- OPEC = classic collusive cartel setting joint output quotas; individual members (e.g., smaller producers) periodically cheat by over-pumping — cartel discipline breaks down periodically.
- Indian telecom price wars (Reliance Jio's 2016 entry) — non-collusive, rivals matched Jio's cuts almost immediately (inelastic-below logic) but wouldn't have matched a price hike (elastic-above logic) -> price rigidity around the "new normal" price point.

Quick-Fire FAQ

Q: What's the single defining trait of oligopoly vs. other structures?

A: Mutual interdependence — a firm cannot set price/output without considering rivals' reactions.

Q: Why does the kinked demand curve predict price stability despite cost changes?

A: Because MR is discontinuous at the kink — moderate MC shifts stay inside the gap, leaving profit-max P^* and Q^* unchanged.

Write in notes: the market structures table (very common short-answer question) and the two-branch elasticity logic behind the kink.

6. Game Strategy (Game Theory)

Core Concept

A formal way to analyze decisions where payoffs depend on what others do (interdependence, as in oligopoly).

- **Elements:** Players (decision-makers), Strategies (available actions), Payoffs (outcomes/utility for each strategy combination), Information sets (what each player knows when deciding).

- **Cooperative games:** binding agreements possible (players can commit/enforce, e.g., cartel with legal contract).
- Non-cooperative games:** no binding agreements — each player acts independently in self-interest (most oligopoly pricing situations).

Decision Rules

- **Dominant strategy:** a strategy that gives a player the best payoff *regardless* of what the other player does. Simplest to identify — check each strategy against both of rival's possible moves.
- **Nash Equilibrium:** a strategy combination where **no player can improve their payoff by unilaterally changing their own strategy**, given the other player's strategy is fixed. Every dominant-strategy equilibrium is a Nash equilibrium, but Nash equilibria can exist without any dominant strategy.

The Prisoner's Dilemma (canonical payoff matrix)

	Player B: Stay Silent	Player B: Confess
Player A: Stay Silent	(-1, -1)	(-10, 0)
Player A: Confess	(0, -10)	(-5, -5)

Format: (Player A payoff, Player B payoff), years in prison as negative (lower magnitude = better).

- **Dominant strategy for both:** Confess (regardless of what the other does, confessing gives a strictly better payoff than staying silent).
- **Nash Equilibrium:** (Confess, Confess) -> (-5, -5).
- **The "dilemma":** (-5,-5) is worse for BOTH than (-1,-1) [mutual silence], but mutual silence isn't stable — each player can unilaterally do better by confessing, so rational self-interest leads to a jointly worse outcome.

Business Application — Duopoly Advertising War

	Firm Y: Low Ad Spend	Firm Y: High Ad Spend
Firm X: Low Ad Spend	(Rs 50cr, Rs 50cr)	(Rs 20cr, Rs 60cr)
Firm X: High Ad Spend	(Rs 60cr, Rs 20cr)	(Rs 30cr, Rs 30cr)

- Each firm's dominant strategy: High Ad Spend (better regardless of rival's choice: 60>50 and 30>20).
- Nash Equilibrium: (High, High) -> (Rs 30cr, Rs 30cr) — both spend more, profits are lower than if both had held back at (Rs 50cr, Rs 50cr). Structurally identical to the Prisoner's Dilemma.
- Same logic applies to **capacity addition races** (both firms over-invest in capacity to avoid being underprepared, eroding industry profitability) and **trade-tariff retaliation** (both countries impose tariffs defensively even though mutual free trade would be jointly better).

Quick-Fire FAQ

Q: Dominant strategy vs. Nash equilibrium — key difference?

A: Dominant strategy is optimal against *every* possible rival move; Nash equilibrium only requires optimality *given the rival's actual chosen strategy* (no incentive to deviate unilaterally). A game can have a Nash equilibrium with no dominant strategy at all.

Q: Why is the Prisoner's Dilemma "used" everywhere in business strategy?

A: Because real duopoly situations (advertising, capacity, pricing, tariffs) share its core structure: individually rational choices -> collectively worse-off equilibrium.

Write in notes: the definitions of dominant strategy and Nash equilibrium verbatim, and practice filling one payoff matrix from scratch — this is the most common numerical question type.

7. Price Discrimination & Advanced Pricing Strategies

Core Concept

Objective: convert **consumer surplus** into **producer surplus** by charging different prices/terms to different units, customers, or times rather than one uniform price.

The Three Classic Degrees

Degree	Mechanism	DWL	Real Example
First (Perfect)	Charge each unit/each customer their exact reservation price	Eliminated (fully captures CS)	Rare in pure form; approximated by personalized bargaining (car dealership haggling, B2B custom quotes)
Second	Non-linear pricing — quantity discounts/block pricing; customer self-selects a tier	Reduced but not zero	Electricity slab tariffs, bulk SKU discounts (buy 3 get 1 free)
Third	Segment market by observable trait with differing elasticities	Reduced but not zero	Student discounts, airline business vs. economy fares

Third-degree rule: charge a **lower price to the more elastic segment**, higher price to the less elastic (inelastic) segment — this maximizes total revenue extraction across segments. Formally, at profit max: $P_i \times (1 - 1/|e_i|)$ is equalized to MC across segments, so the group with higher elasticity magnitude (e) gets the lower price.

Specialized Pricing Tactics

- **Intertemporal PD (skimming):** Launch high, cut price over time as high-willingness-to-pay early adopters are captured first, then price drops to capture the rest. E.g., new smartphones (iPhone launch price $\rightarrow$ falls over its life cycle); hardcover books priced high at launch, paperback (lower price) released later for price-sensitive buyers.
- **Peak-load pricing:** Price varies with **time-variant marginal cost of capacity**, not just demand — during peak periods, serving one more customer requires expensive capacity expansion, so peak price is set higher (partly cost-based, partly surplus extraction). E.g., surge pricing on ride-hailing apps, higher electricity tariffs during peak evening hours.
- **Two-part tariff:** Fixed access fee (extracts CS, ideally set near the lowest-valuation consumer's total surplus) + a per-unit variable price (often set closer to MC to keep usage efficient). E.g., theme park entry fee + per-ride charge (classic Disneyland-pricing problem); gym membership fee + per-class charge; club membership.
- **Bundling:** *Pure bundling* — only sold as a package (no a la carte). *Mixed bundling* — both bundle and individual items available. Works best when customer valuations for different products are **negatively correlated** (one loves product A but not B, another the reverse) — bundling averages out the variance in willingness-to-pay and lets the seller capture more surplus than either pure component pricing or pure bundling alone. E.g., OTT platform bundles (sports + movies + originals in one subscription), bank combo accounts (savings + insurance + demat bundled).

Graph

First-Degree (Perfect) Price Discrimination

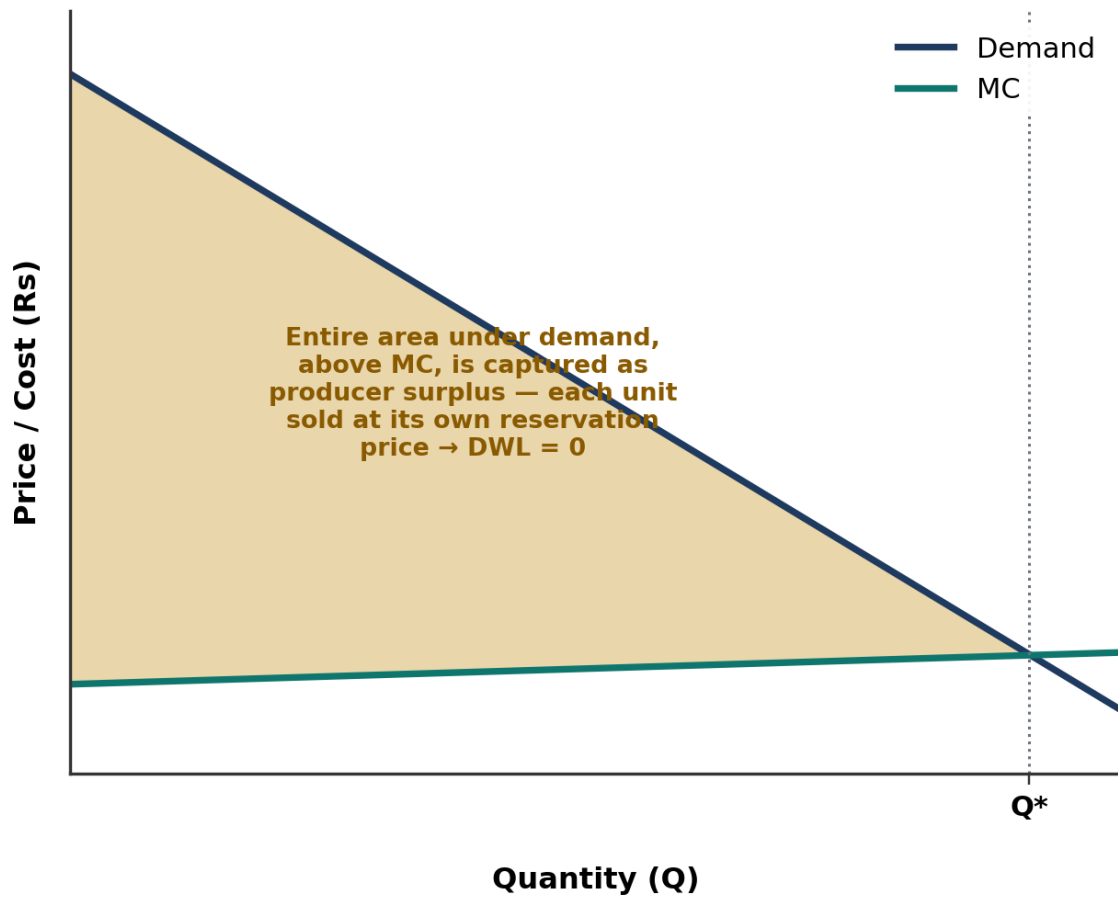
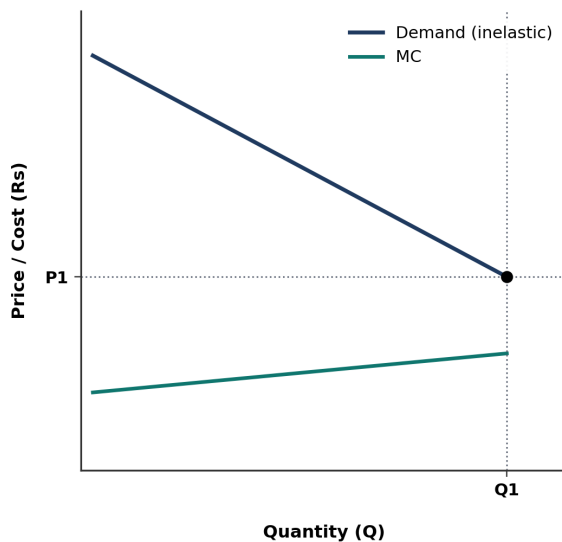


Figure 7.1 — First-Degree (Perfect) Price Discrimination

Third-Degree Price Discrimination — Segmenting by Elasticity

Inelastic Segment (e.g. Business Travel)
 $\rightarrow$ charge the HIGHER price



Elastic Segment (e.g. Leisure/Student)
 $\rightarrow$ charge the LOWER price

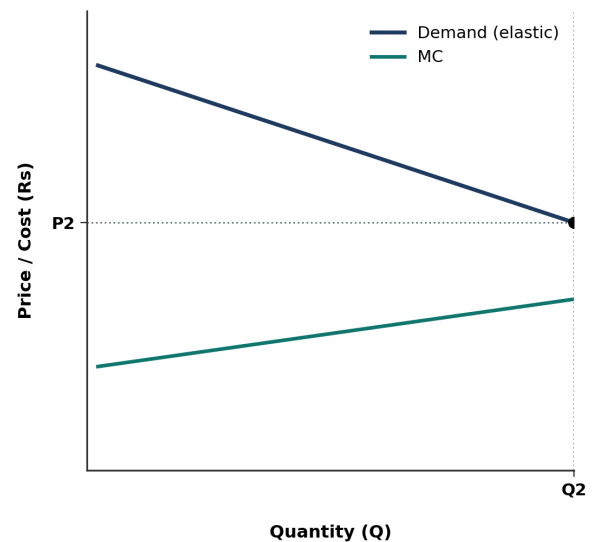


Figure 7.2 — Third-Degree Price Discrimination: Segmenting by Elasticity

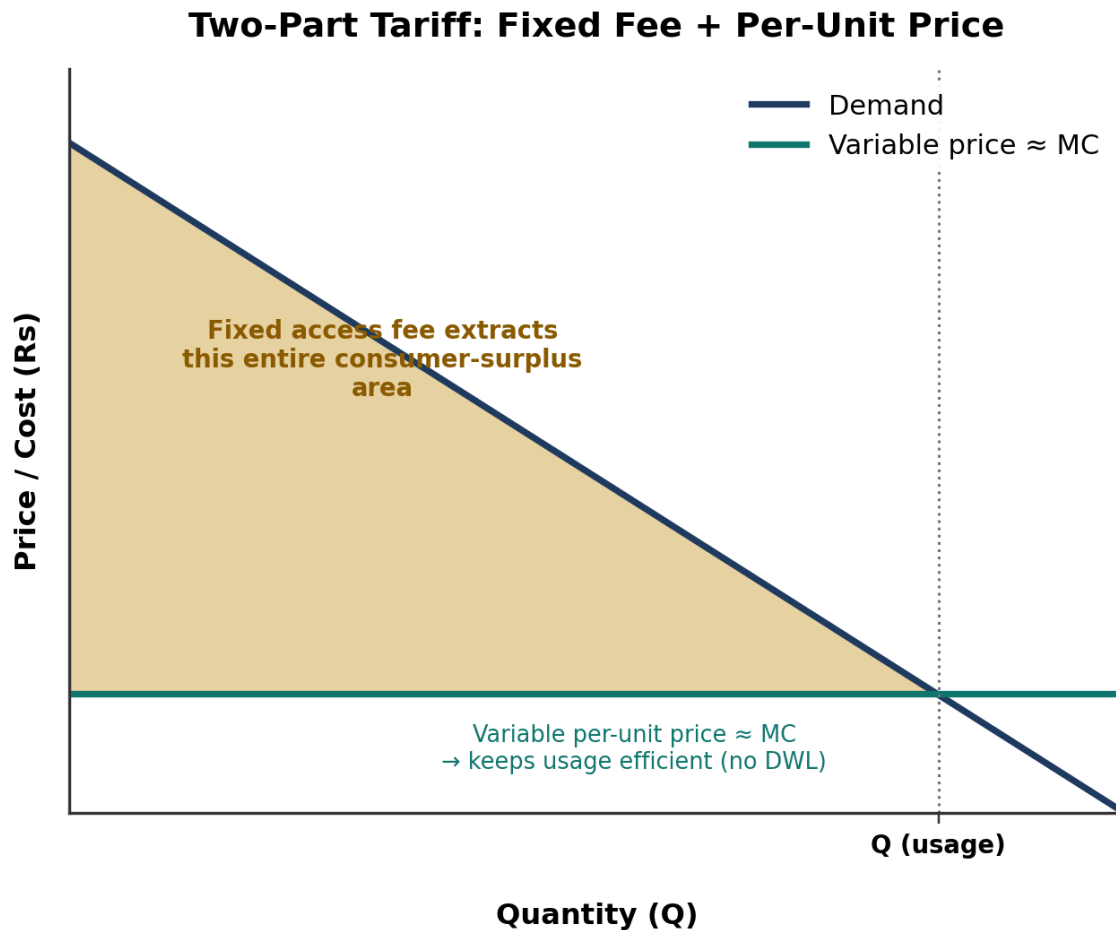


Figure 7.3 — Two-Part Tariff: Fixed Fee + Per-Unit Price

Quick-Fire FAQ

Q: Why lower price to the elastic segment in third-degree PD?

A: Elastic buyers are highly price-sensitive — a small price cut brings in disproportionately more volume/revenue from them, while inelastic buyers will pay more without much volume loss, so profit-maximization equalizes marginal revenue per segment, not price.

Q: When does bundling work best?

A: When individual product valuations are negatively correlated across customers — it smooths out variance and lets the seller price closer to average total valuation.

Q: Two-part tariff: why set variable price near MC?

A: To avoid discouraging efficient usage (deadweight loss from overpricing units); the fixed fee is where most surplus extraction happens.

Write in notes: the third-degree "lower price to elastic segment" rule and the negative-correlation condition for bundling — both are frequently tested conceptually, not just numerically.

8. Sustainability & Market Failures

Core Concept

- **Externality:** a cost or benefit from production/consumption that falls on a third party NOT involved in the transaction, and is therefore not reflected in market price.
- **Negative externality (production):** e.g., industrial effluent discharge harms downstream communities/ecosystems, cost not borne by the polluting firm.
- **Negative externality (consumption):** e.g., single-use plastic disposal creates landfill/ocean pollution costs borne by society, not the consumer.

Decision Rule / Math

- **Marginal Private Cost (MPC):** cost borne directly by the producer (what shows up in their supply curve).

$$\text{MSC (Marginal Social Cost)} = \text{MPC} + \text{MEC (Marginal External Cost)}$$

Market equilibrium (based on $\text{MPC} = \text{Demand}$) sets output at Q_{market} , but the **socially efficient** output is where $\text{MSC} = \text{Demand}$, at $Q_{\text{social}} < Q_{\text{market}}$.

Because the market ignores MEC, it **overproduces** relative to the social optimum -> deadweight loss (DWL) from unpriced externality.

Graph

Negative Externality: Private vs Social Cost

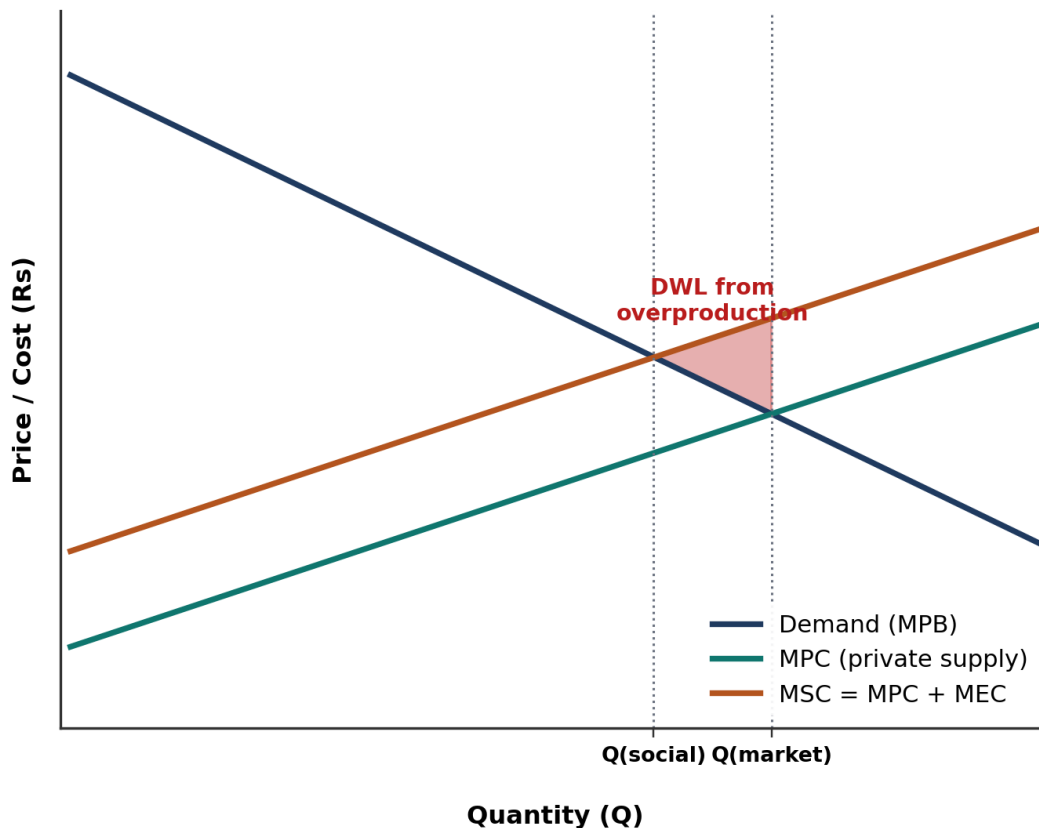


Figure 8.1 — Negative Externality: Private vs Social Cost and the Resulting DWL

Methods to Internalize Externalities

Method	Mechanism	Example
Pigouvian tax	Tax = MEC at optimal Q, forces firm to internalize the external cost, shifts MPC up to meet MSC	Carbon tax, plastic cess
Cap-and-trade / tradable permits	Government caps total allowable pollution, firms trade permits — price discovery lets low-cost abaters sell to high-cost abaters	EU ETS carbon credits
Circular business models	Redesign production to reduce/reuse waste at the source rather than pricing the externality after the fact	Closed-loop packaging (refillable bottles)
Extended Producer Responsibility (EPR)	Legally makes producers responsible for a product's post-consumer disposal/recycling cost	E-waste/battery take-back mandates in India

Real-World Example

A textile dyeing unit dumping untreated effluent into a river: MPC (its own dyeing cost) is lower than MSC (which includes downstream water-treatment and health costs) -> market overproduces dyed fabric relative to the social optimum until a Pigouvian effluent tax or EPR-style mandate forces internalization.

Quick-Fire FAQ

Q: Why does an unpriced negative externality cause overproduction, not underproduction?

A: Because the firm only weighs its own (lower) MPC against demand, ignoring the extra MEC borne by society — so it keeps producing past the point where true social cost exceeds social benefit.

Q: Pigouvian tax vs. cap-and-trade — key operational difference?

A: Pigouvian tax fixes the *price* of pollution (quantity floats); cap-and-trade fixes the *quantity* of pollution (price floats, discovered via permit trading).

Write in notes: $MSC = MPC + MEC$ and the direction of the DWL triangle — this is the single most-repeated externality diagram across exams.

That's the full syllabus. Want practice/numerical problems (payoff matrix fill-ins, $MR=a-2bQ$ calculations, shutdown-rule scenarios) for drilling any of these eight topics?